(APT) Behavior From the Void: Unsupervised Active Pre-Training

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- Related Works
  - Unsupervised Reinforcement Learning
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  - Mutual Information Maximization
  - Contrastive Learning
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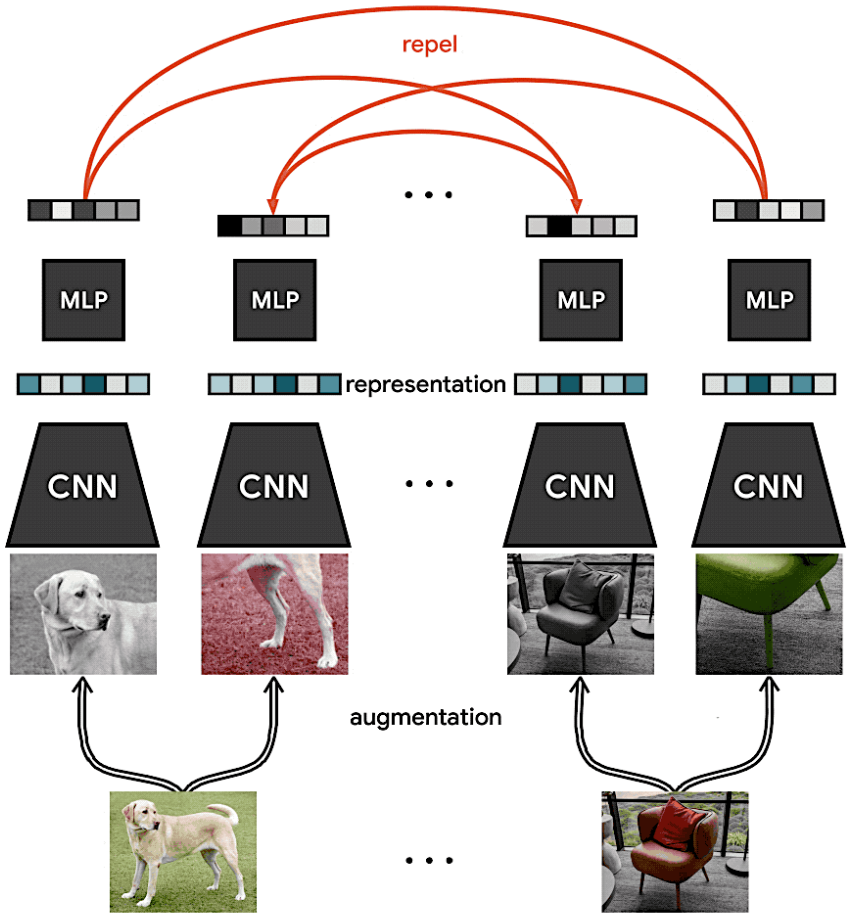
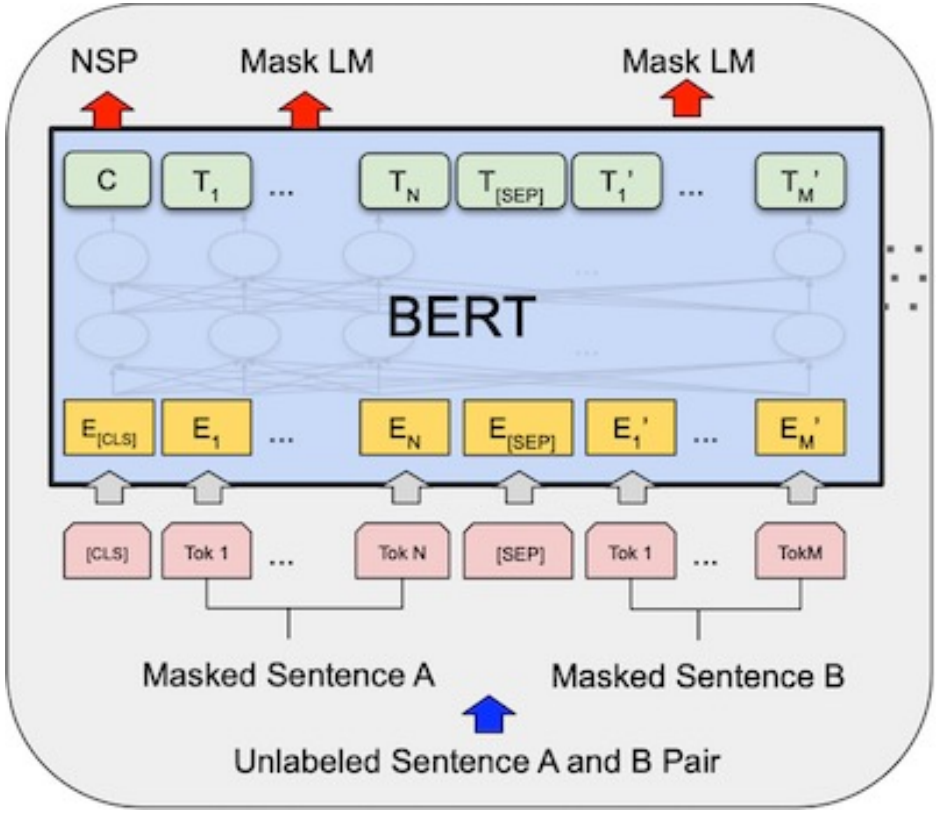
Success story of Unsupervised Pre-training

- Unsupervised pre-training has obtained promising results in CV and NLP.



SimCLR

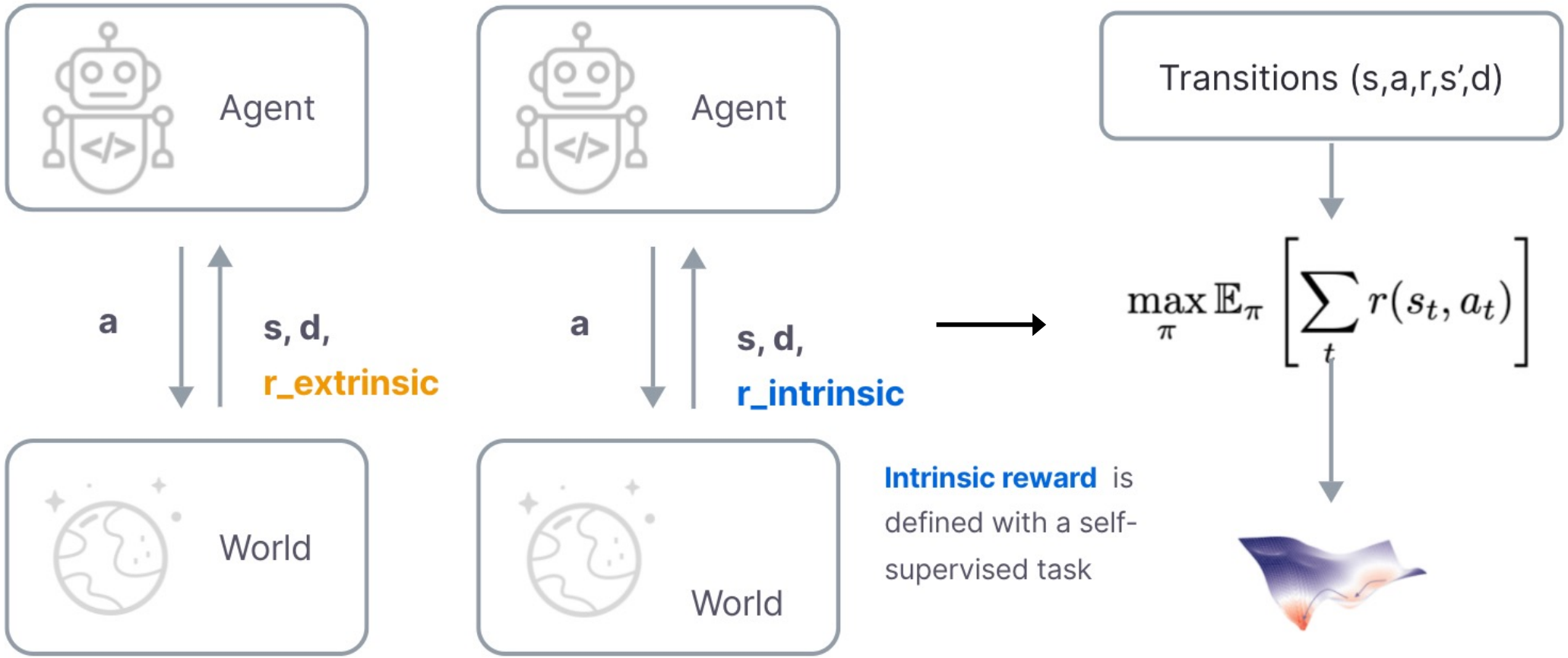


BERT

[1] A simple framework for contrastive learning of visual representations., Chen et al., ICML, 2020.
[2] BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding., Devlin et al., NAACL, 2019.

Unsupervised Pre-training in Reinforcement Learning

- Supervised RL
- Unsupervised RL
- Finetuning



State (s), Action (a), Reward (r), Done (d)

(Preliminary) DIAYN: Diversity is all you need

- **DIAYN**: maximize the <u>mutual information</u> between policy-conditioned latent (w) and state (s)
  - $\max I(s;w) = \max -H(w|s) + H(w)$
  - $\mathrm{I}(\mathrm{s};w) \geq E_{s,w}[\underline{q_\theta(w|s)} - \underline{\log p(w)}]$

variational approximation

policy latent should be distinguishable

policy latent should be diverse

  - $\mathrm{r}_{\mathrm{intrinsic}}(\mathrm{s},\mathrm{a}) = q_\theta(w|s) - \log p(w)$
  - policy: $\pi(a|s,w)$

[1] Diversity is all you need., Eysenbach et al., ICLR, 2018.

[2] Explore, Discover and Learn: Unsupervised Discovery of State-Covering Skills., Campos et al., ICML, 2020.

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- **Problem: Pessimistic Exploration**
  - Agent asymptotically receives larger rewards for visiting known states than discovering unknown states as proven in [2].

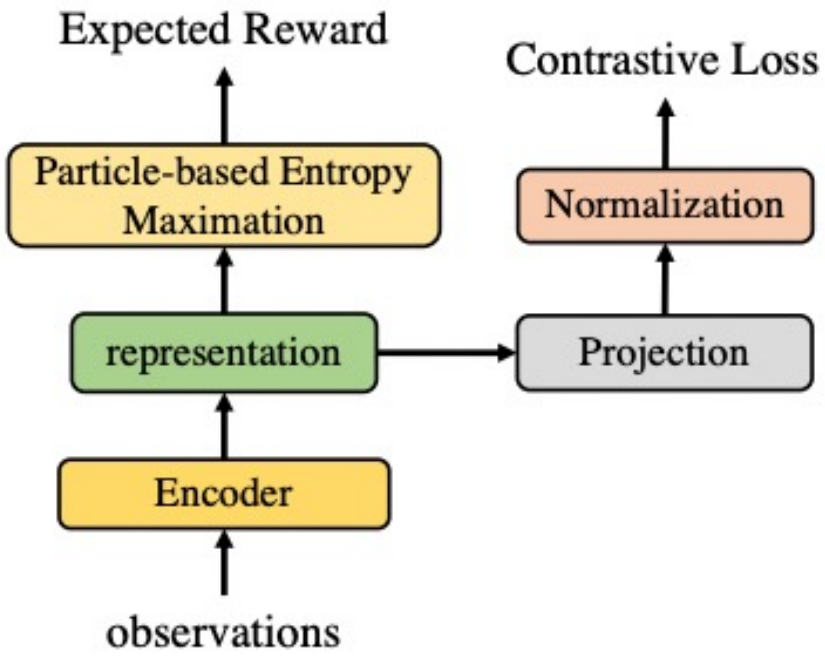
[1] Diversity is all you need., Eysenbach et al., ICLR, 2018.

[2] Explore, Discover and Learn: Unsupervised Discovery of State-Covering Skills., Campos et al., ICML, 2020.

APT: Active Pre-Training for RL

- To overcome the pessimistic exploration problem, incentivize the agent with a reward to <u>maximize entropy</u> in an abstract representation space.

- Overview



- The contrastive loss learns a mapping from state space to an abstract representation space.

- Entropy of states are measured by particle-based estimator and used it as an intrinsic reward.

$$r_{intrinsic} = \text{entropy}(s)$$

APT: Active Pre-Training for RL

- Particle-based Entropy Estimation
  - Assume we have number of n data points $\{z\}_{i=1}^{n}$ from unknown distribution.
  - $H_{particle}(z) = -\frac{1}{n}\sum_{i=1}^{n} \log \frac{k}{n v_i^k} + b(k)$

$$\propto \sum_{i=1}^{n} \log v_i^k$$

$$\propto \sum_{i=1}^{n} \log \underline{\| z_i - z_i^k \|^{n_z}}$$

k nearest neighbor of z_i

$$v_i^k = \frac{\| z_i - z_i^k \|^{n_z} \pi^{n_z/2}}{\Gamma(n_z + 1)}$$

Sparsity of distribution can be measured by the distance between each sampled point and its k nearest neighbor

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  - Assume we have number of n data points $\{z\}_{i=1}^{n}$ from unknown distribution.
  - $H_{particle}(z) = -\frac{1}{n}\sum_{i=1}^{n}\log\frac{k}{nv_i^k} + b(k)$

$$\propto \sum_{i=1}^{n}\log v_i^k$$

$$\propto \sum_{i=1}^{n}\log \| z_i - z_i^k \|^{n_z}$$

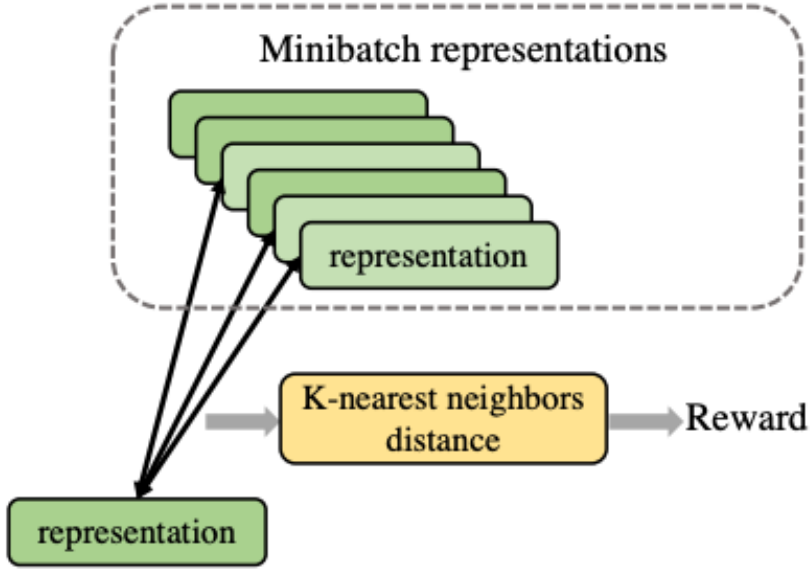
k nearest neighbor of z_i

$$v_i^k = \frac{\| z_i - z_i^k \|^{n_z}\ \pi^{n_z/2}}{\Gamma(n_z+1)}$$

$$:= \sum_{i=1}^{n}\log\left(\frac{1}{k}\sum_{j\in N_k(z_i)} \| z_i - z_i^j \|^{n_z}\right)$$

Average the distance over all k nearest neighbor for stability

  - $r_{intrinsic}(s,a) = \log\frac{1}{k}\sum_{j\in N_k(f_\theta(s))} \| f_\theta(s) - z_i^j \|^{n_z})$



APT: Active Pre-Training for RL

- Learning Contrastive Representation
  - Identical to the SimCLR objective
    - $s_i^k = aug(s_i), \;\; s_i^v = aug(s_i)$
    - $$\min_{\theta,\phi} -\frac{1}{2n}\sum_{i=1}^{n}\left[\log \frac{\exp(h_\phi(f_\theta(s_i^k))^T h_\phi(f_\theta(s_i^v)))}{\sum_{i=1}^{n}\mathbb{I}_{[j\neq i]}(\exp(h_\phi(f_\theta(s_i^k))^T h_\phi(f_\theta(s_j^k))) + \exp(h_\phi(f_\theta(s_i^k))^T h_\phi(f_\theta(s_j^v))))}\right]$$

APT: Active Pre-Training for RL

- Algorithm

Algorithm 1: Training APT

Randomly Initialize f encoder
Randomly Initialize π and Q networks
for $e := 1, \infty$ **do**
  for $t := 1, T$ **do**
    Receive observation s_t from environment
    Take action $a_t \sim \pi(\cdot|s_t)$, receive observation s_{t+1} and $\cancel{r_t}$ from environment
    $\mathcal{D} \leftarrow \mathcal{D} \cup (s_t, a_t, \cancel{r_t}, s'_t)$
    $\{(s_i, a_i, \cancel{r_i}, s'_i)\}_{i=1}^{N} \sim \mathcal{D}$ `// sample a mini batch`
    Train neural encoder f on mini batch `// representation learning`
    for *each* $i = 1..N$ **do**
      $a'_i \sim \pi(\cdot|s'_i)$
      $\hat{Q}_i = Q_{\theta'}(s'_i, a'_i)$
      Compute r_{APT} with equation (5) `// particle-based entropy reward`
      $y_i \leftarrow r_{\text{APT}} + \gamma \hat{Q}_i$
    end
    $loss_Q = \sum_i (Q(s_i, a_i) - y_i)^2$
    Gradient descent step on Q and π `// standard actor-critic`
  end
end

Experiments

- ATARI benchmark (Pretrain: 250M, Finetune: 100k)
  - Comparison with Unsupervised RL baselines

Table 3: Evaluation in Atari games. The amount of RL interaction utilized is 100K. Mdn is the median of human-normalized scores, M is the mean and $> H$ is the number of games with human-level performance. CL denotes training representation encoder using contrastive learning and data augmentation. On each subset, we mark as bold the highest score.

	26 Game Subset			Full 57 Games		
Algorithm	Mdn	M	$> H$	Mdn	M	$> H$
CBB	1.23	21.94	3	–	–	–
MEPOL	0.34	17.94	2	–	–	–
DIAYN	1.34	25.39	2	2.95	23.90	6
CBB w/ CL	1.78	17.34	2	–	–	–
MEPOL w/ CL	1.05	21.78	3	–	–	–
DIAYN w/ CL	1.76	28.44	2	3.28	25.14	6
VISR	9.50	**128.07**	**7**	6.81	**102.31**	11
APT w/o CL	21.23	28.12	3	28.65	41.12	9
APT	**47.50**	69.55	**7**	**33.41**	47.73	**12**

All baseline models benefit from contrastive represenatiton learning

Experiments

- ATARI benchmark (Pretrain: 250M, Finetune: 100k)
  - Comparison with Sample Efficient RL baselines

Game	Random	Human	SimPLe	DER	CURL	DrQ	SPR	VISR	APT (ours)
Alien	227.8	7127.7	616,9	739.9	558.2	771.2	801.5	364.4	**2614.8**
Amidar	5.8	1719.5	88.0	188.6	142.1	102.8	176.3	186.0	**211.5**
Assault	222.4	742.0	527.2	431.2	600.6	452.4	571.0	**12091.1**	891.5
Asterix	210.0	8503.3	1128.3	470.8	734.5	603.5	977.8	**6216.7**	185.5
Bank Heist	14.2	753.1	34.2	51.0	131.6	168.9	380.9	71.3	**416.7**
BattleZone	2360.0	37187.5	5184.4	10124.6	14870.0	12954.0	**16651.0**	7072.7	7065.1
Boxing	0.1	12.1	9.1	0.2	1.2	6.0	**35.8**	13.4	21.3
Breakout	1.7	30.5	16.4	1.9	4.9	16.1	17.1	**17.9**	10.9
ChopperCommand	811.0	7387.8	**1246.9**	861.8	1058.5	780.3	974.8	800.8	317.0
Crazy Climber	10780.5	23829.4	**62583.6**	16185.2	12146.5	20516.5	42923.6	49373.9	44128.0
Demon Attack	107805	35829.4	62583.6	16185.3	12146.5	20516.5	42923.6	**8994.9**	5071.8
Freeway	0.0	29.6	20.3	27.9	26.7	9.8	24.4	-12.1	**29.9**
Frostbite	65.2	4334.7	254.7	866.8	1181.3	331.1	**1821.5**	230.9	1796.1
Gopher	257.6	2412.5	771.0	349.5	669.3	636.3	715.2	498.6	**2590.4**
Hero	1027.0	30826.4	2656.6	6857.0	6279.3	3736.3	**7019.2**	663.5	6789.1
Jamesbond	29.0	302.8	125.3	301.6	471.0	236.0	365.4	**484.4**	356.1
Kangaroo	52.0	3035.0	323.1	779.3	872.5	940.6	**3276.4**	1761.9	412.0
Krull	1598.0	2665.5	**4539.9**	2851.5	4229.6	4018.1	2688.9	3142.5	2312.0
Kung Fu Master	258.5	22736.3	17257.2	14346.1	14307.8	9111.0	13192.7	16754.9	**17357.0**
Ms Pacman	307.3	6951.6	1480.0	1204.1	1465.5	960.5	1313.2	558.5	**2827.1**
Pong	-20.7	14.6	**12.8**	-19.3	-16.5	-8.5	-5.9	-26.2	-8.0
Private Eye	24.9	69571.3	58.3	97.8	218.4	-13.6	**124.0**	98.3	96.1
Qbert	163.9	13455.0	1288.8	1152.9	1042.4	854.4	669.1	666.3	**17671.2**
Road Runner	11.5	7845.0	5640.6	9600.0	5661.0	8895.1	**14220.5**	6146.7	4782.1
Seaquest	68.4	42054.7	683.3	354.1	384.5	301.2	583.1	706.6	**2116.7**
Up N Down	533.4	11693.2	3350.3	2877.4	2955.2	3180.8	**28138.5**	10037.6	8289.4
Mean HNS	0.000	1.000	44.3	28.5	38.1	35.7	**70.4**	64.31	69.55
Median HNS	0.000	1.000	14.4	16.1	17.5	26.8	41.5	12.36	**47.50**
# Superhuman	0	N/A	2	2	2	2	**7**	6	**7**

Experiments

- Mujoco Benchmark (Pretrain: 5M, Finetune: 100k)
  - Comparison with Sample Efficient RL baselines

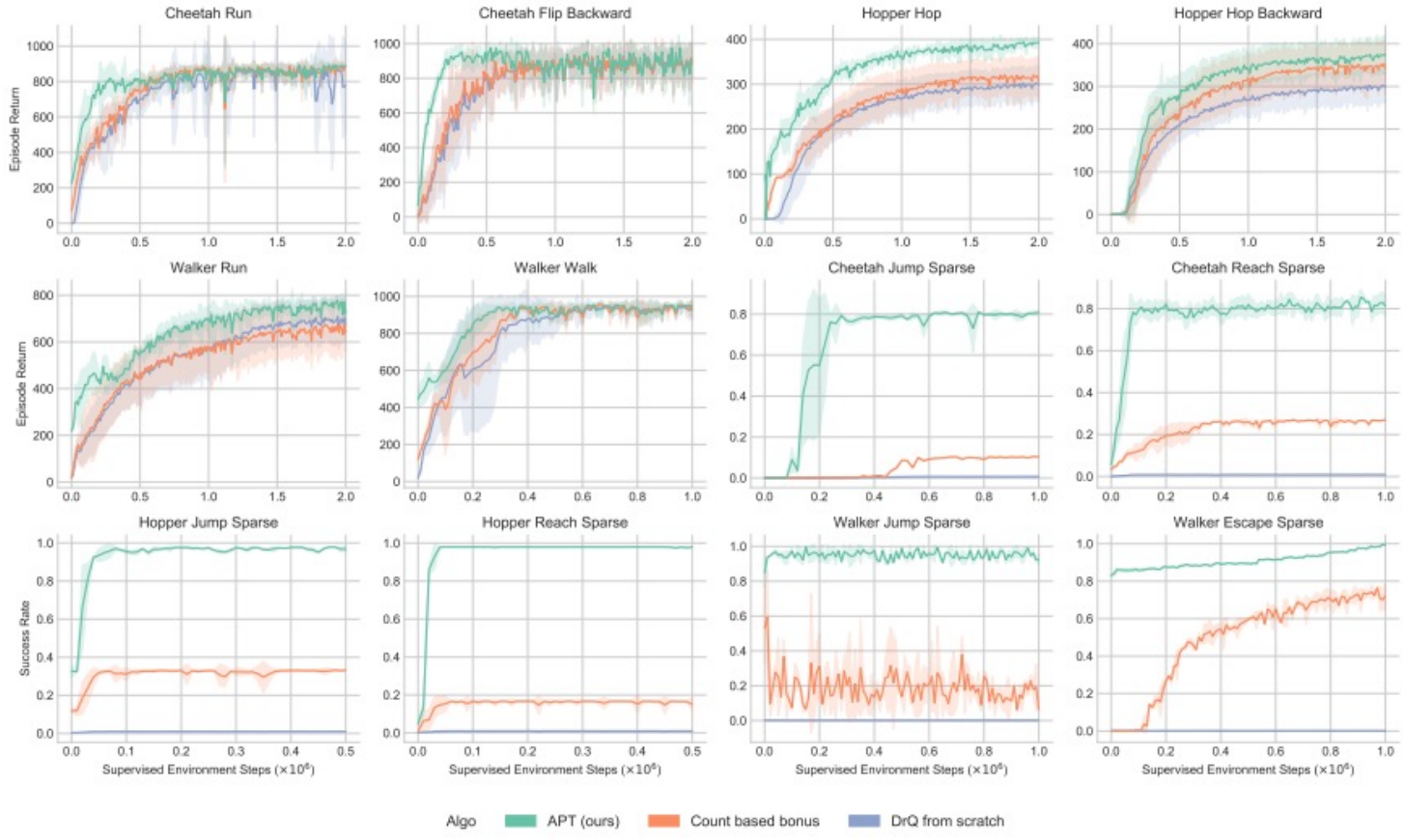


Figure 3: Results of different methods in environments from DMControl. All curves are the average of three runs with different seeds, and the shaded areas are standard errors of the mean.

Limitations

- Exhaustive Pre-training time (250M)

  → Pre-training time should be restricted to a realistic number? (10M…?)

- Is online pre-training realistic?

  → more realistic evaluation protocol for unsupervised reinforcement learning might be…

  1. pre-train the model with offline data
  2. evaluate with reward-based finetuning.

- Representation Learning with 3-layer ConvNet?

  → Does the performance scales robustly with model size?

Q & A